

Problema 29.1.

(a) Utilizati interpolarea Hermite pentru a gasi un polinom de grad minim ce satisface

$$p(-1) = p'(-1) = 0, p(0) = 1, p(1) = p'(1) = 0.$$

Simplificati expresia lui p cat mai mult posibil.

$$\begin{array}{lll} x_0 = -1 & r_0 = -1 & f(x_0) = f'(x_0) = 0 \\ x_1 = 0 & r_1 = 0 & f(x_1) = 1 \\ x_2 = 1 & r_2 = 1 & f(x_2) = f'(x_2) = 0 \end{array}$$

$$m = 2 \quad n = m + r_0 + r_1 + r_2 = 4 \Rightarrow \text{gradul maxim al polinomului este } 4$$

Polinomul de interpolare al lui Hermite se calculeaza cu formula:

$$(H_n f)(x) = \sum_{i=0}^m \sum_{l=0}^{r_i} f^{(l)}(x_i) h_{il}(x), \text{ unde:}$$

$$h_{ir_i}(x) = \frac{(x-x_i)^{r_i}}{r_i!} q_i(x) \quad \text{si} \quad h_{il}(x) = \frac{(x-x_i)^l}{l!} q_i(x) - \sum_{j=l+1}^{r_i} \binom{r_i}{j} q_i^{j-l}(x_i) h_{ij}(x)$$

pentru $l = \overline{0, r_i - 1}$,

$$q_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^m \left(\frac{x-x_j}{x_i-x_j} \right)^{r_j+1}$$

Astfel avem:

$$q_0(x) = \prod_{\substack{j=0 \\ j \neq 0}}^2 \left(\frac{x-x_j}{x_0-x_j} \right)^{r_j+1} = -x \frac{(x-1)^2}{4}, \quad q_1(x) = \prod_{\substack{j=0 \\ j \neq 1}}^2 \left(\frac{x-x_j}{x_1-x_j} \right)^{r_j+1} = (x-1)^2 (x+1)^2$$

$$q_2(x) = \prod_{\substack{j=0 \\ j \neq 2}}^2 \left(\frac{x-x_j}{x_2-x_j} \right)^{r_j+1} = x \frac{(x+1)^2}{4}$$

$$h_{01}(x) = \frac{(x-x_0)^{r_0}}{r_0!} q_0(x) = -\frac{x(x+1)(x-1)^2}{4} \quad h_{10}(x) = \frac{(x-x_1)^{r_1}}{r_1!} q_1(x) = (x-1)^2(x+1)^2$$

$$h_{21}(x) = \frac{(x-x_2)^{r_2}}{r_2!} q_2(x) = \frac{x(x-1)(x+1)^2}{4}$$

$$h_{00}(x) = \frac{(x-x_0)^0}{0!} q_0(x) - \sum_{j=1}^0 \binom{j}{0} q_0^j(x_0) h_{0j}(x) = q_0(x) - h_{01}(x)$$

$$h_{20}(x) = \frac{(x-x_2)^0}{2!} q_2(x) - \sum_{j=1}^1 \binom{j}{0} q_2^j(x_0) h_{2j}(x) = \frac{q_2(x)}{2} - h_{21}(x)$$

=>

$$(H_n f)(x) = f(x_0) * h_{00}(x) + h_{01}(x) * f'(x_0) + h_{10}(x) * f(x_1) + \\ + f(x_2) * h_{20}(x) + h_{21}(x) * f'(x_2)$$

$$\Rightarrow (H_n f)(x) = h_{10}(x) * f(x_1) = (x+1)^2(x-1)^2$$

(b) Presupunem ca polinomul p de la (a) este utilizat pentru a aproxima functia

$$f(x) = [\cos(\Pi x/2)]^2 \text{ pe } -1 \leq x \leq 1$$

(b1) Exprimati eroarea $(Rf)(x) = f(x) - p(x)$ (pentru un x din $[-1, 1]$ fixat) in functie de o derivata corespunzatoare a lui f .

$$(Rf)(x) = \frac{1}{(m+1)!} \prod_{i=0}^m (x-x_i)^{r_i} f^{(n+1)}(x)$$

$$(Rf)(x) = \frac{1}{3!} \prod_{i=0}^2 (x-x_i)^{r_i} f^{(5)}(x)$$

$$f^{(5)}(x) = -\frac{\Pi^5}{2} \sin(\Pi x)$$

$$\Rightarrow (Rf)(x) = -\frac{\Pi^5}{12} (x-1)(x+1) \sin(\Pi x)$$

$$\text{Pentru } x = \frac{1}{2} \text{ se obtine } (Rf)\left(\frac{1}{2}\right) = \frac{3\Pi^5}{48}$$

(b2) Gasiti o margine superioara a lui $|(Rf)(x)|$ (pentru un $x \in [-1, 1]$ fixat).

$$|(R_n f)(x)| \leq \frac{M_{n+1}}{(n+1)!} |u_n(x)|, \text{ unde:}$$

$$M_{n+1}=\max_{x\in[-1,1]}|f^{(n+1)}(x)|\quad \text{si } u_n(x)=\prod_{i=0}^m(x-x_i)^{r_i+1}$$

$$M_5=\max_{x\in[-1,1]}|-\frac{\Pi^5}{2}\sin(\Pi x)|=\frac{\Pi^5}{2}\qquad u_4(x)=x(x+1)^2(x-1)^2$$

$$\Rightarrow |(R_nf)(x)|\leq \frac{\Pi^5}{240}|x|(x+1)^2(x-1)^2$$

$$x=\frac{1}{2}\quad \Rightarrow |(R_nf)(x)|\leq \frac{3\Pi^5}{2560}$$